GEOGRAPHY

PHYSICAL GEOGRAPHY

GEOMORPHOLOGY

MODULE - 31

FLUVIAL LANDFORMS

- PROCESS OF RIVER ACTION
- STAGES OF RIVER

FLUVIAL LANDFORMS

THE DEVELOPMENT OF RIVER SYSTEM

1 Denudation is the general lowering of Earth's surface by Geomorphic agents like river, ice wind and waves.

2 Unlike glacier and snow which are confined to the temperate and cold latitudes, waves which acts only on coast lines, winds which are only efficient in deserts, the effect of running water is felt all over the globe, wherever water is present.

3 The source of a river may be a spring or a lake or a marsh, which is generally present in upland region, where precipitation is heaviest and where there is a slope downwards through which the run off can flow.

FLUVIAL LANDFORMS

THE DEVELOPMENT OF RIVER SYSTEM

3 The uplands which receive the heaviest precipitation and contributes the greatest amount of water for the flow of rivers, therefore, form the ***catchment areas of rivers.***

4 The crest of the mountains is the ***divide or watershed*** from which streams flow down the slopes on both sides to begin their journey to the oceans.

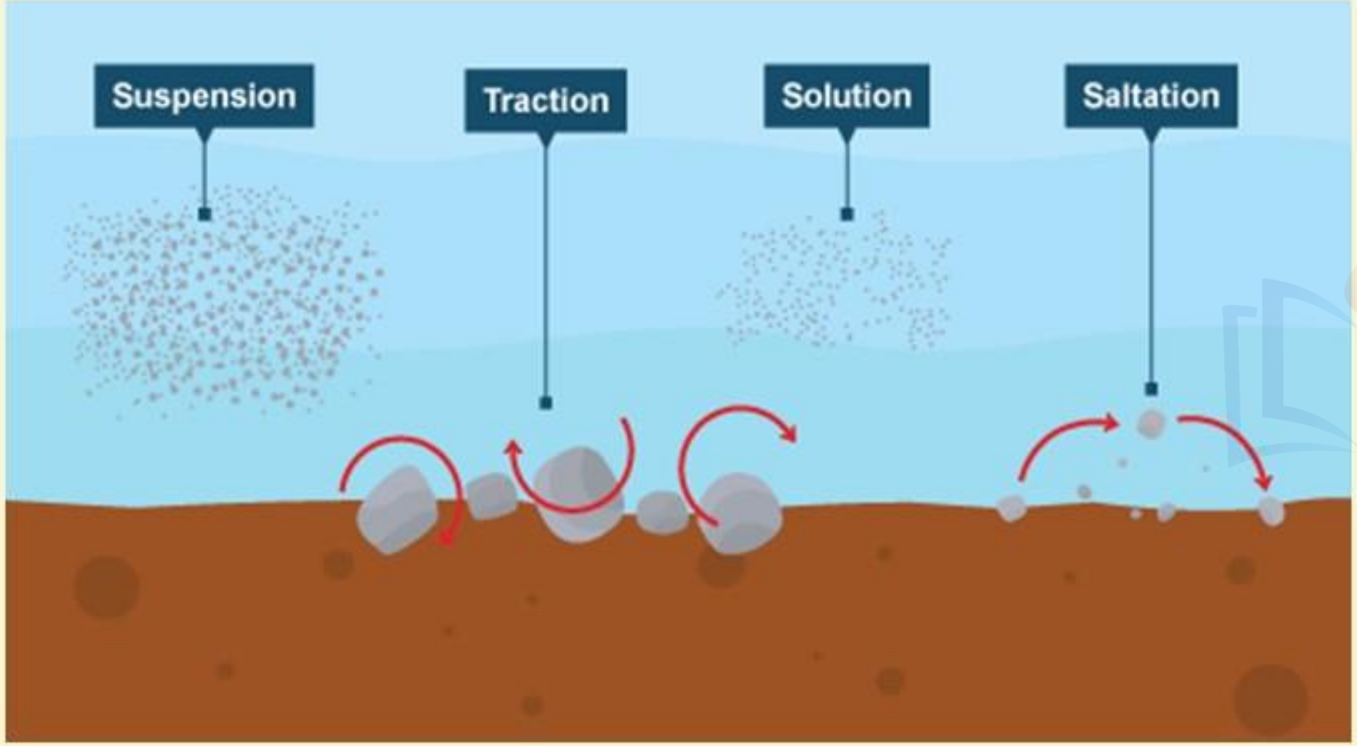
FLUVIAL LANDFORMS

PROCESS OF RIVER EROSION AND TRANSPORTATION

1. Mass movements of Earth on hill slopes is mainly due to the lubricating action of water which allows a mass of material to move under gravity.
2. This is particularly acute where the slopes are steep. The slow motion of soil down the hill slope is called soil creep; a more sudden movement due to the lubricating effect of water may cause landslides.

FLUVIAL LANDFORMS

PROCESS OF RIVER EROSION AND TRANSPORTATION



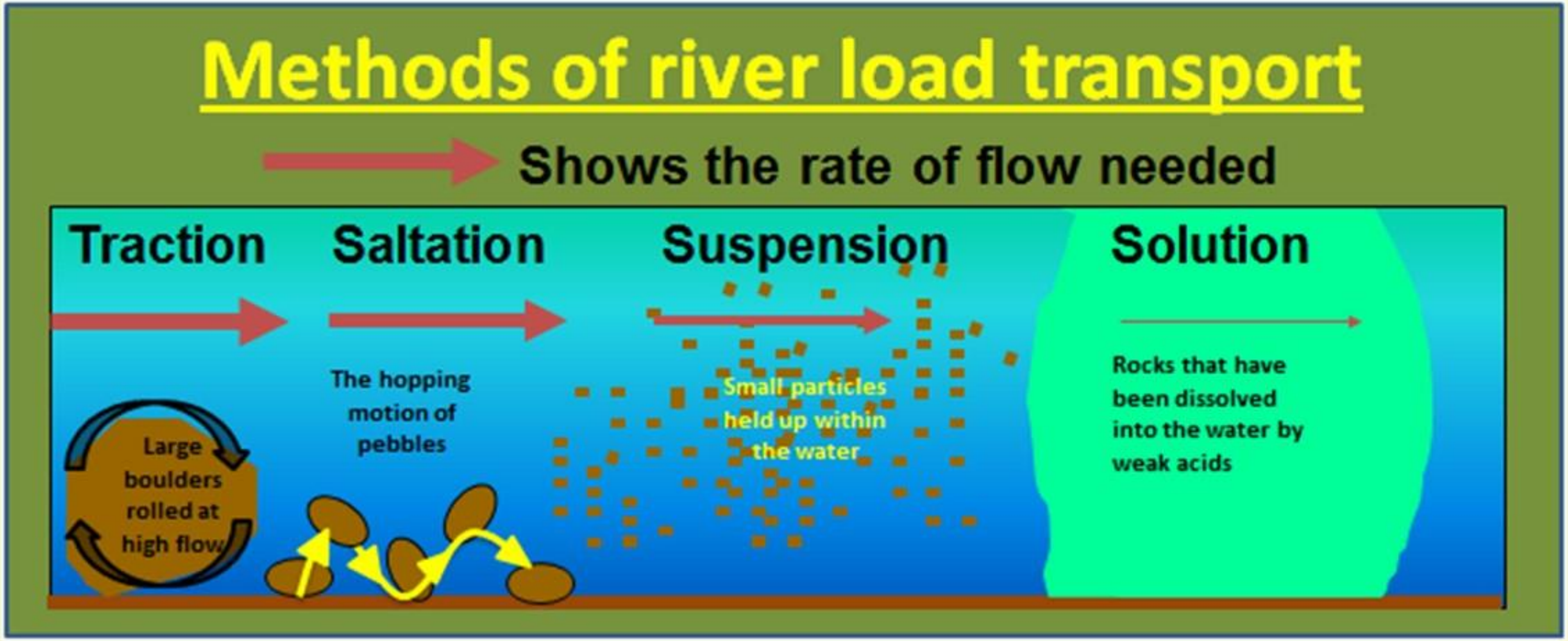
1 When a river flows, it carries with it eroded materials. These comprise of the river load and may be divided into three types:

1) Materials in Solution- These are minerals which are dissolved in the water.
2) Materials in Suspension- Sand, silt and mud are carried along suspended in the river stream
3) The Traction Load- This includes coarser materials such as pebbles, stones and boulders, which are rolled along the river beds.

The Mississippi river which drains an area almost half of the size of the United States removes more than 2 million tonnes of eroded material into the Gulf of Mexico daily. Consequently the river basins are lowered.

FLUVIAL LANDFORMS

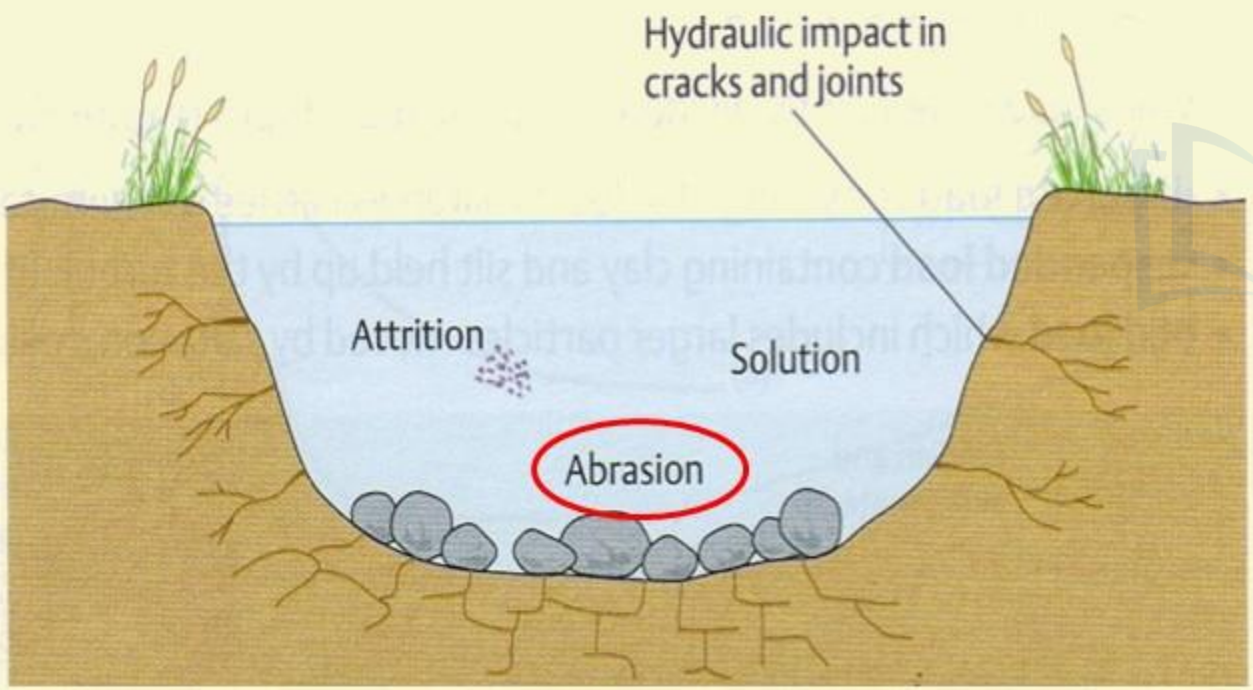
PROCESS OF RIVER EROSION AND TRANSPORTATION



FLUVIAL LANDFORMS

PROCESS OF RIVER EROSION AND TRANSPORTATION

The process of erosion and transportation in rivers go on simultaneously comprising the following process:



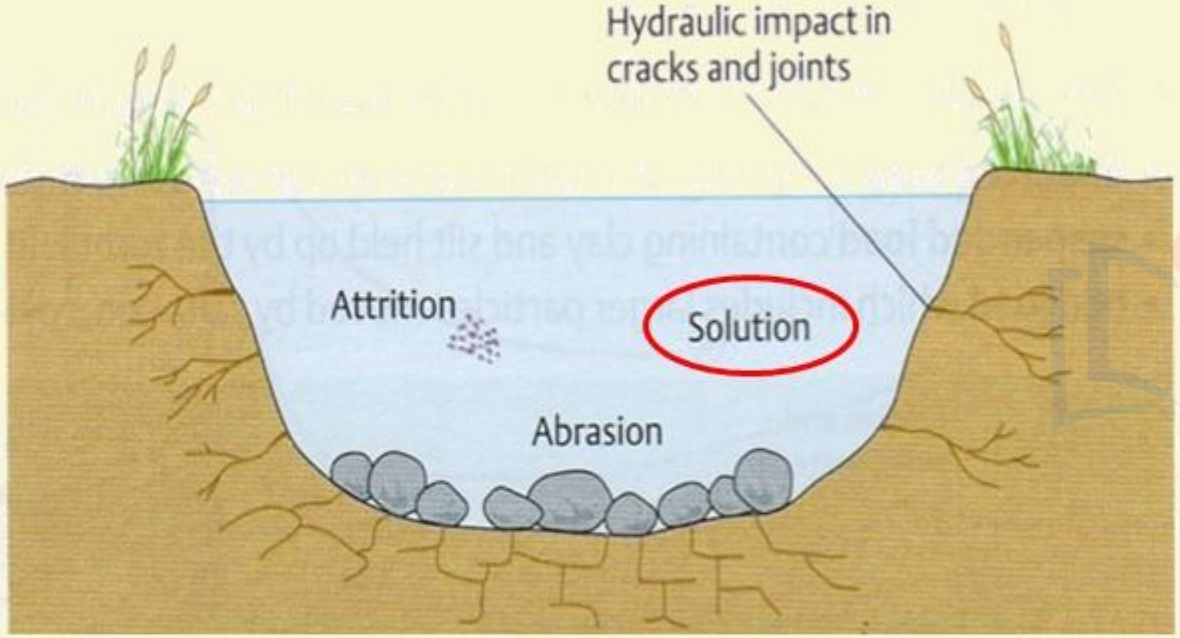
1 Corrasion or Abrasion:

This is the mechanical grinding of the river's traction load against the banks and bed of the river. The rock fragments are hurled against the sides of the river and also roll along the bottom of the river. Corrasion takes place, in two distinct ways.

1. Lateral corrasion- sideways erosion which widens the V-shaped valley.
2. Vertical corrasion- the downward action which deepens the river channel.

FLUVIAL LANDFORMS

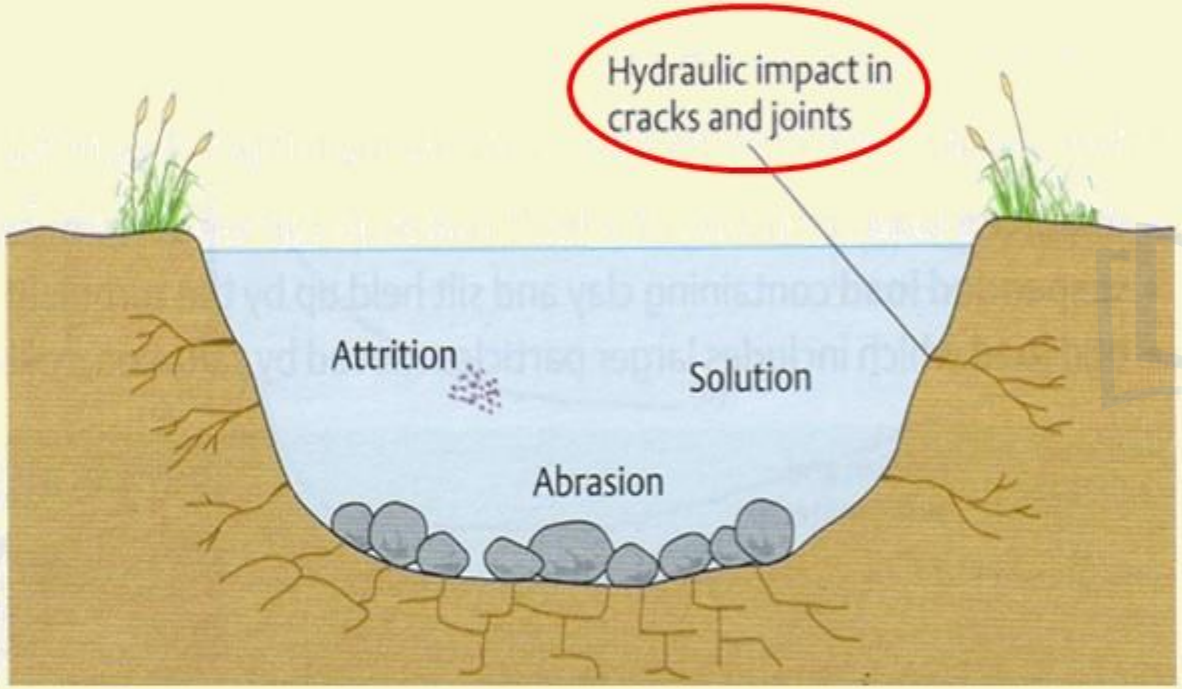
PROCESS OF RIVER EROSION AND TRANSPORTATION



2 **Corrosion or solution:**
This is the chemical action of water on the soluble rocks with which the river comes into contact. For example calcium carbonate in limestone's is easily dissolved and removed in solution.

FLUVIAL LANDFORMS

PROCESS OF RIVER EROSION AND TRANSPORTATION

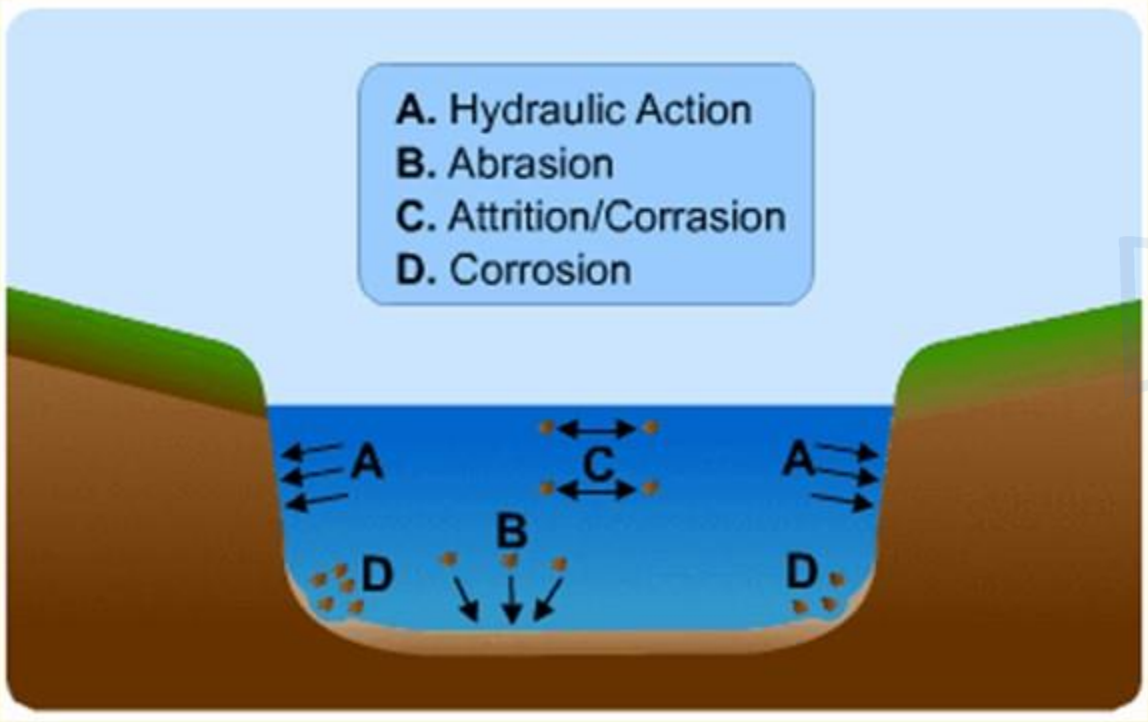


3 Hydraulic action:

- This is the mechanical loosening and sweeping away of materials by the river water itself.
- Some of the water splashes against the river banks and surges into cracks and crevices.
- This helps to disintegrate the rocks. The water also undermines the softer rocks with which it comes into contact.

FLUVIAL LANDFORMS

PROCESS OF RIVER EROSION AND TRANSPORTATION



4 Attrition:

- This is the wear and tear of the transported materials themselves when they roll and collide into one another.
- The coarser boulders are broken down into smaller stones, the angular edges are smoothed and rounded to form pebbles.
- The finer materials are carried further down-stream to be deposited.

FLUVIAL LANDFORMS

THE PROCESS OF RIVER ACTION

1. During the floods the amount of rock debris swept off by river is very greater. We can see this from the colour of the flood water during heavy rain.

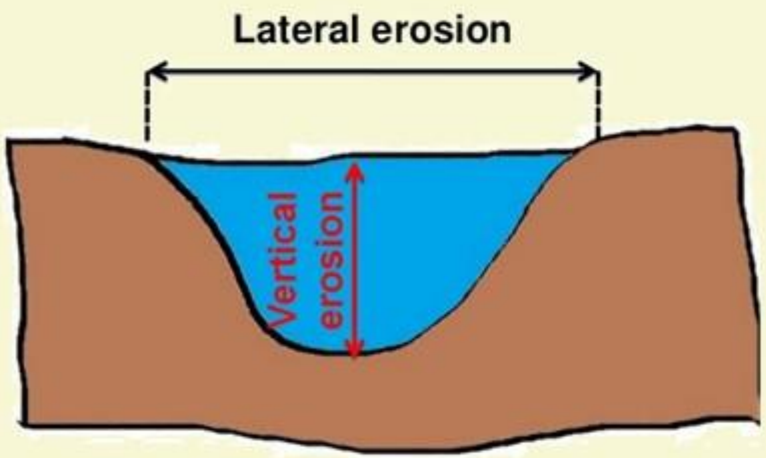
2. The ability of water to move the various materials depends upon ***the volume of water, the velocity of flow and lastly the shape and weight of the load.***

3. It is said that by doubling the velocity of water, its transporting power is increased by more than 10 times. Thus, huge boulders which are stranded in normal times may be moving during seasonal floods.

FLUVIAL LANDFORMS

THE PROCESS OF RIVER ACTION

4 Most of the erosional landforms made by running water are associated with vigorous and youthful rivers flowing along gradients.



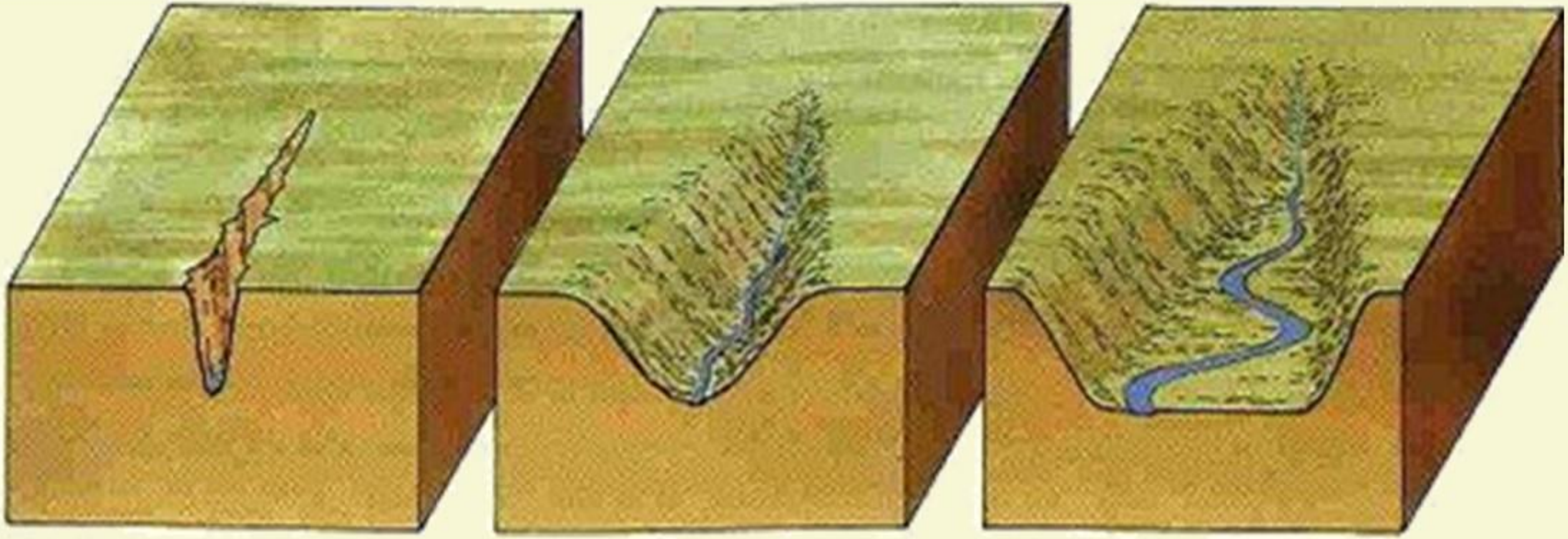
5 With time, stream channels turn gentler (due to continued erosion) and as a consequence streams lose their velocity facilitating active deposition.

6 There may be depositional forms associated with streams flowing over steep slopes. But these phenomena will be on a small scale compared to those associated with rivers flowing over medium to gentle slopes.

FLUVIAL LANDFORMS

THE PROCESS OF RIVER ACTION

7 The gentler the river channel's gradient or slope, the greater is the deposition. When the stream beds turn gentler due to continued erosion, downward cutting becomes less dominant and lateral erosion of banks increases and as a consequence the hills and valleys are reduced to plains.

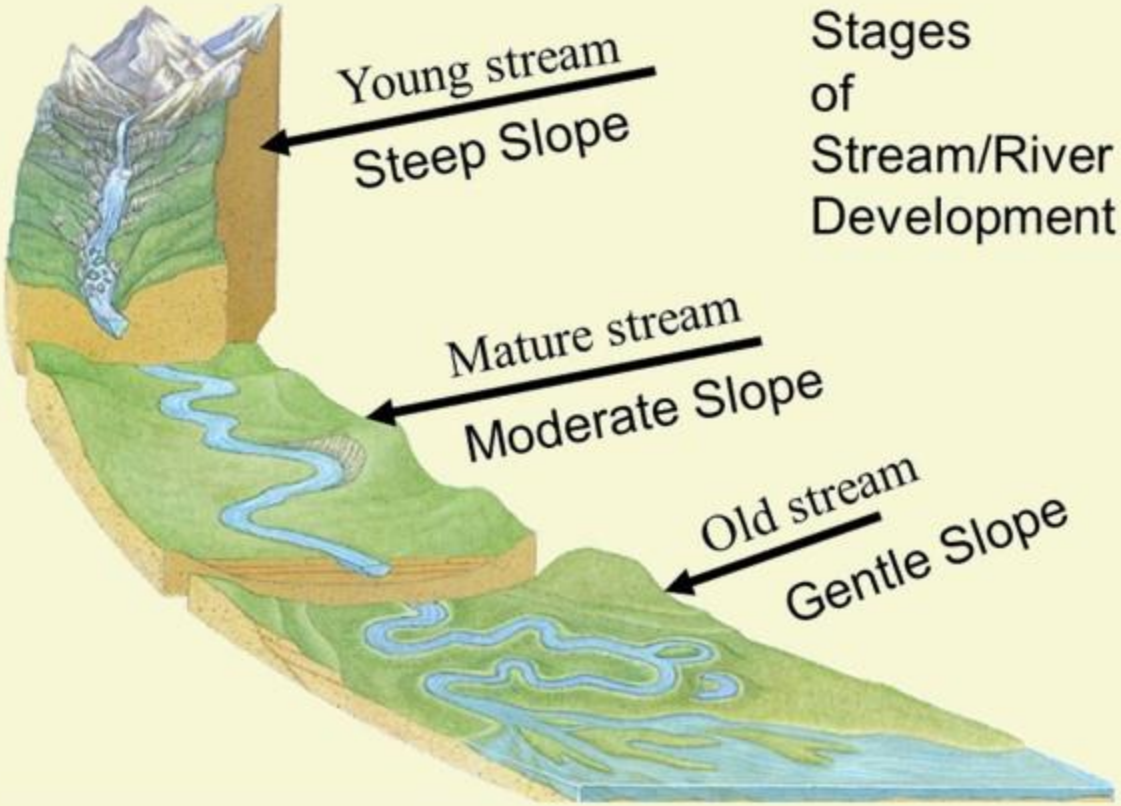


The river cuts down and deepens its valley | **The river widens its valley as it deepens it** | **The river continues to widen its valley**

Learning Space

FLUVIAL LANDFORMS

THE PROCESS OF RIVER ACTION



Stages of Stream/River Development

8 Because of the sheer friction of the column of flowing water, minor or major quantities of materials from the surface of the land are removed in the direction of flow and gradually small and narrow rills will form.

9 These rills will gradually develop into long and wide gullies; the gullies will further deepen, widen, lengthen and unite to give rise to a network of valleys.

10 In the early stages, down-cutting dominates; in the middle stages, streams cut their beds slower, and lateral erosion of valley sides becomes severe. Gradually, the valley sides are reduced to lower and lower slopes.

FLUVIAL LANDFORMS

THE PROCESS OF RIVER ACTION

11 In the old stage, the divides between drainage basins are likewise lowered until they are almost completely flattened leaving finally a lowland with some low resistant remnants called monadnocks standing out here and there.

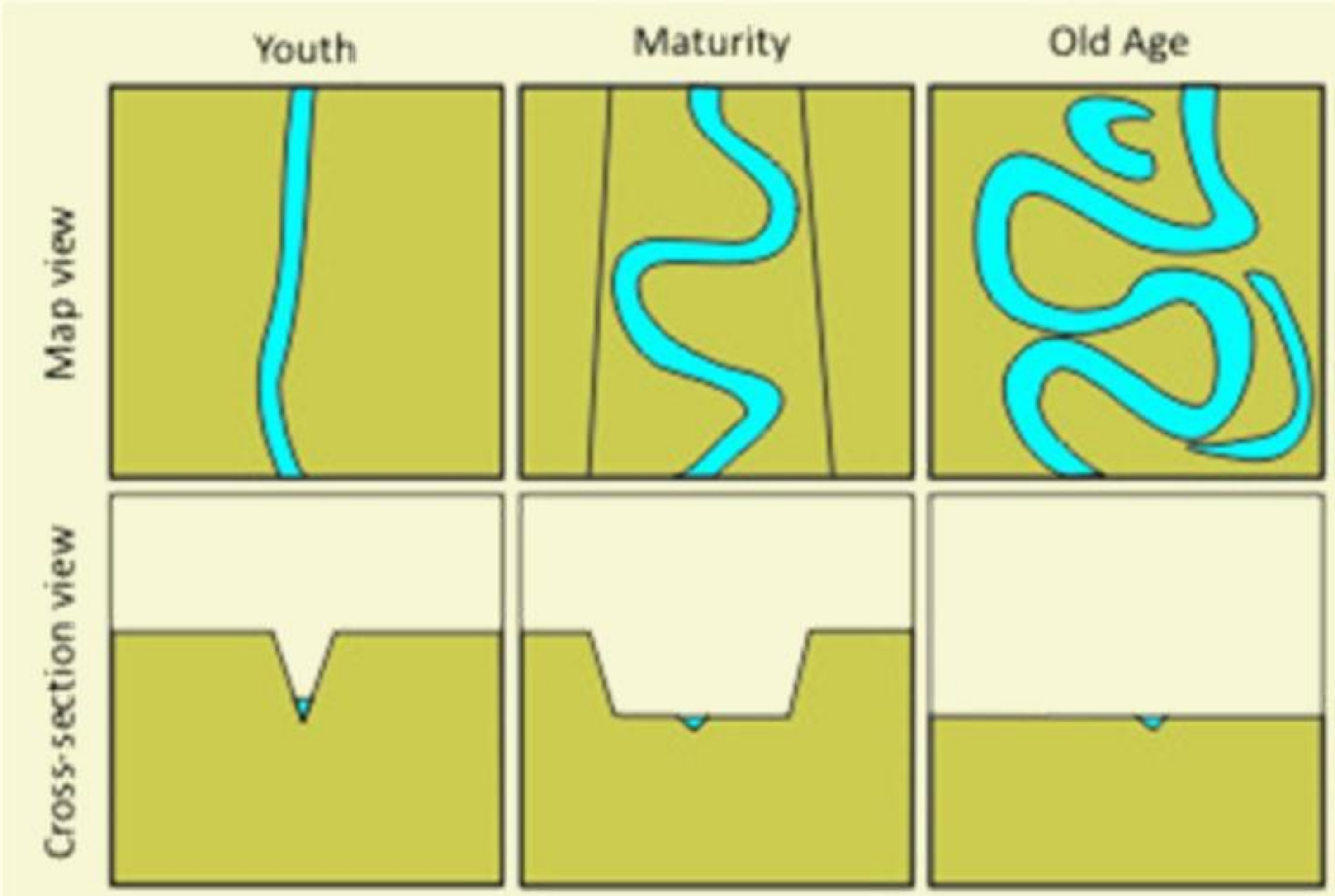
12 This type of plain forming as a result of stream erosion is called a peneplain (an almost plain).

MONADNOCKS

PENEPLAIN

FLUVIAL LANDFORMS

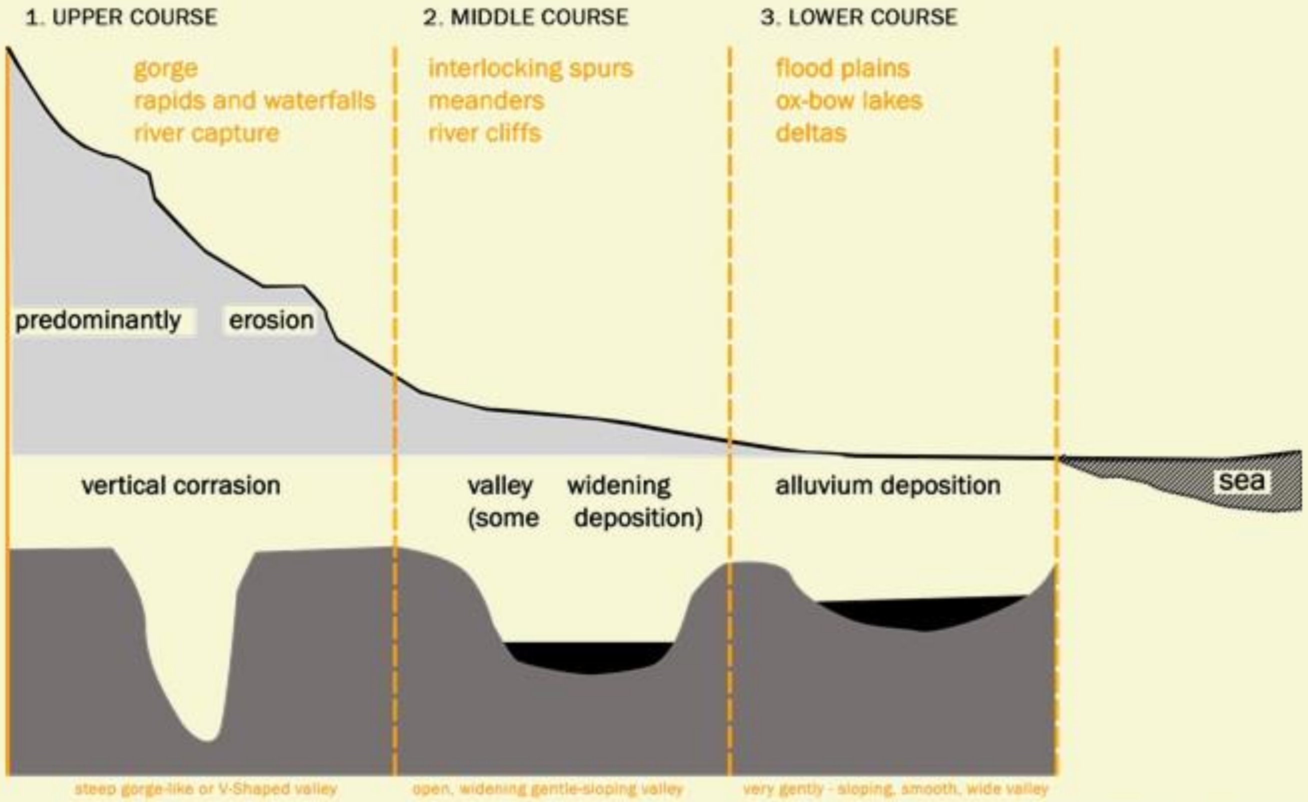
THE PROCESS OF RIVER ACTION



FLUVIAL LANDFORMS

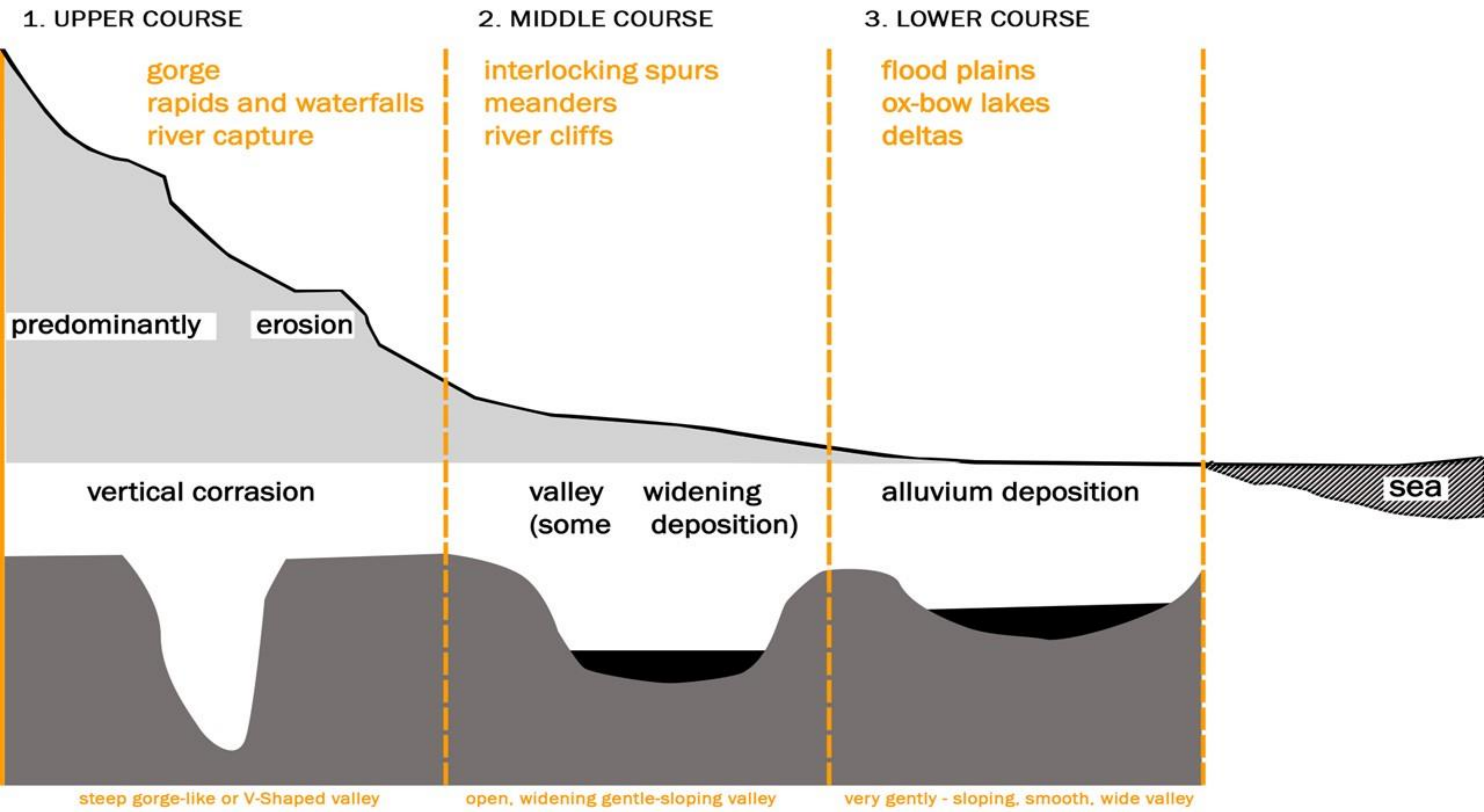
THE COURSE OF RIVER

The characteristics of each of the stages of landscapes developing in running water regimes may be summarized as follows:



The course of a river may be divided into three distinct parts

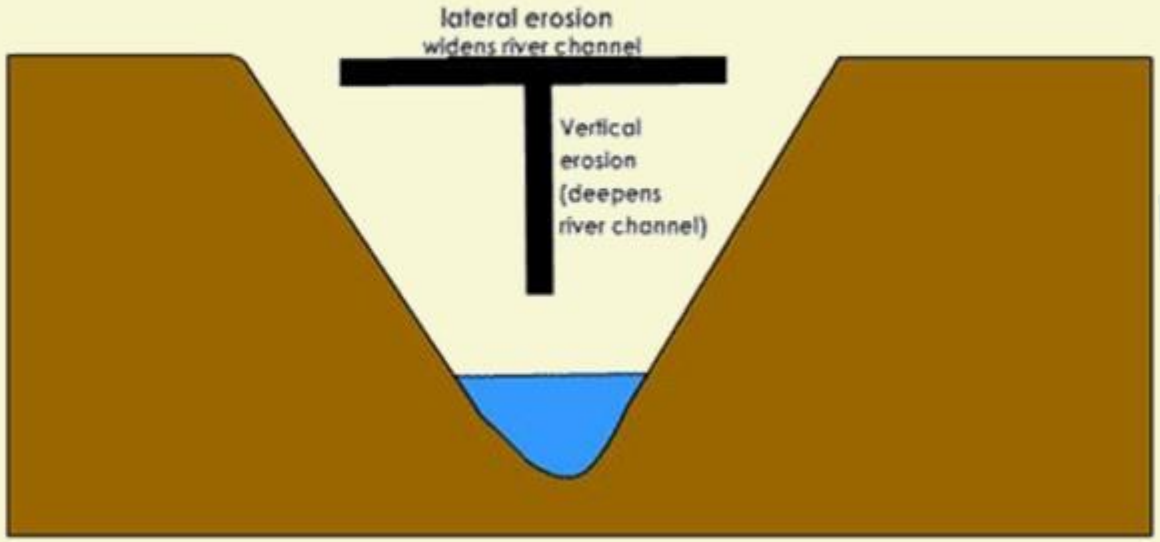
1) The upper or mountain course- the stage of youth,
2) The middle or valley course- the stage of maturity.
3) The lower or plain course- the stage of old age.


1. UPPER COURSE
2. MIDDLE COURSE
3. LOWER COURSE
gorge
rapids and waterfalls
river capture
interlocking spurs
meanders
river cliffs
flood plains
ox-bow lakes
deltas
predominantly erosion
vertical corrasion
valley widening (some deposition)
alluvium deposition
sea
steep gorge-like or V-Shaped valley
open, widening gentle-sloping valley
very gently - sloping, smooth, wide valley

FLUVIAL LANDFORMS

THE COURSE OF RIVER

1 THE UPPER OR MOUNTAIN COURSE



1. This begins at the source of the river near the watershed, which is probably the crest of a mountain range.
2. The river is very swift as it descends the steep slopes, and the predominant action of the river is vertical corrasion.
3. Down-cutting takes place so rapidly that lateral corrasion will be at its lowest in this stage.

FLUVIAL LANDFORMS

THE COURSE OF RIVER

1 THE UPPER OR MOUNTAIN COURSE

4 In some cases where the rocks are very resistant, the valley is narrow and the sides are so steep that gorges are formed e.g. the Indus Gorge in Kashmir.

5 In arid regions, where there is little rainfall to widen the valley sides, and the river cuts deep into the valley-floor, and the valleys so formed is called Canyons.

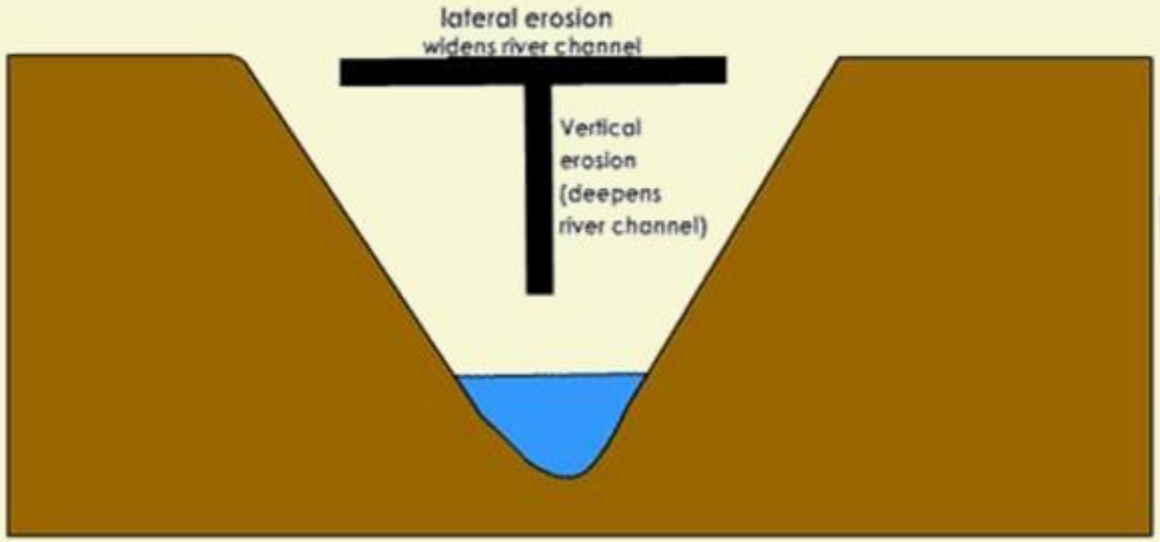
INDUS GORGE

GRAND CANYON

FLUVIAL LANDFORMS

THE COURSE OF RIVER

2 THE MIDDLE OR VALLEY COURSE



1. In the middle course, lateral corrasion tends to replace vertical corrasion.
2. Active erosion of the banks widens the V-shaped valley.
3. The volume of water increases with the confluence of many tributaries and this increases the river's load.
4. The work of the river is predominantly transportation with some deposition.
5. Rain-wash, soil creep and gullying gradually widens the valley, cutting back the sides of the river.

FLUVIAL LANDFORMS

THE COURSE OF RIVER

2 THE MIDDLE OR VALLEY COURSE

MEANDERS

6 The river's task of valley-cutting, bed-smoothing and debris-removal are being carried out in a more tranquil manner than in the mountain course though the velocity does not decrease.

7 Some of the load is dropped or deposited. Again this depends on the volume of the flow.

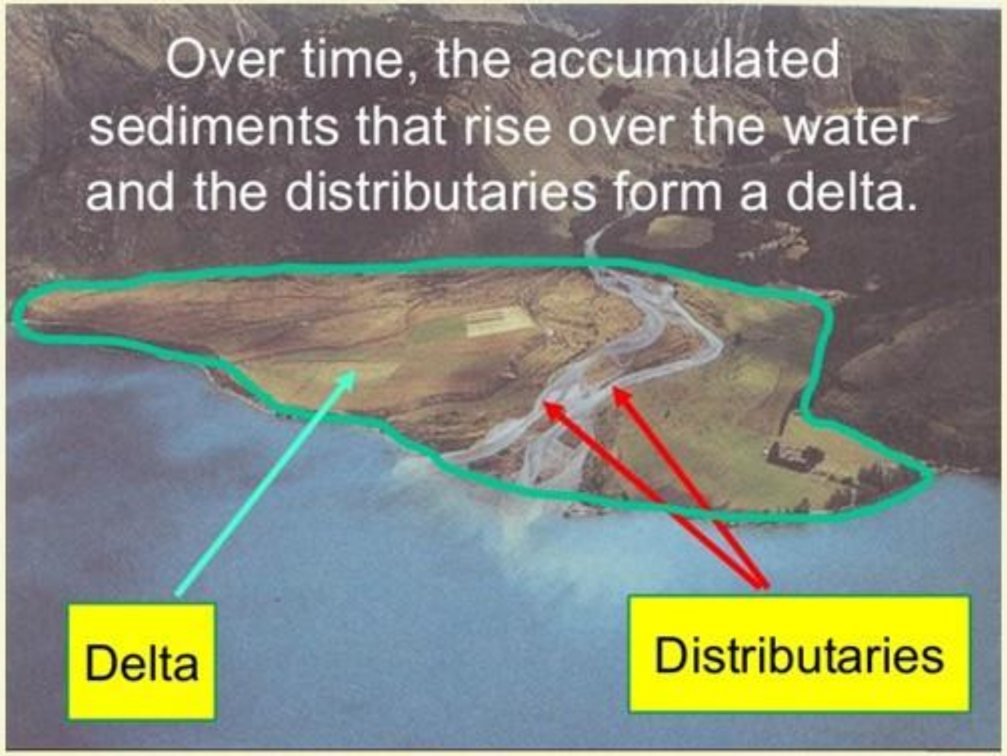
8 The more outstanding features associated with the valley course are the following

a) Meanders
b) River cliffs and slip-off slopes
c) Interlocking spurs

FLUVIAL LANDFORMS

THE COURSE OF RIVER

3 THE LOWER OR PLAIN COURSE

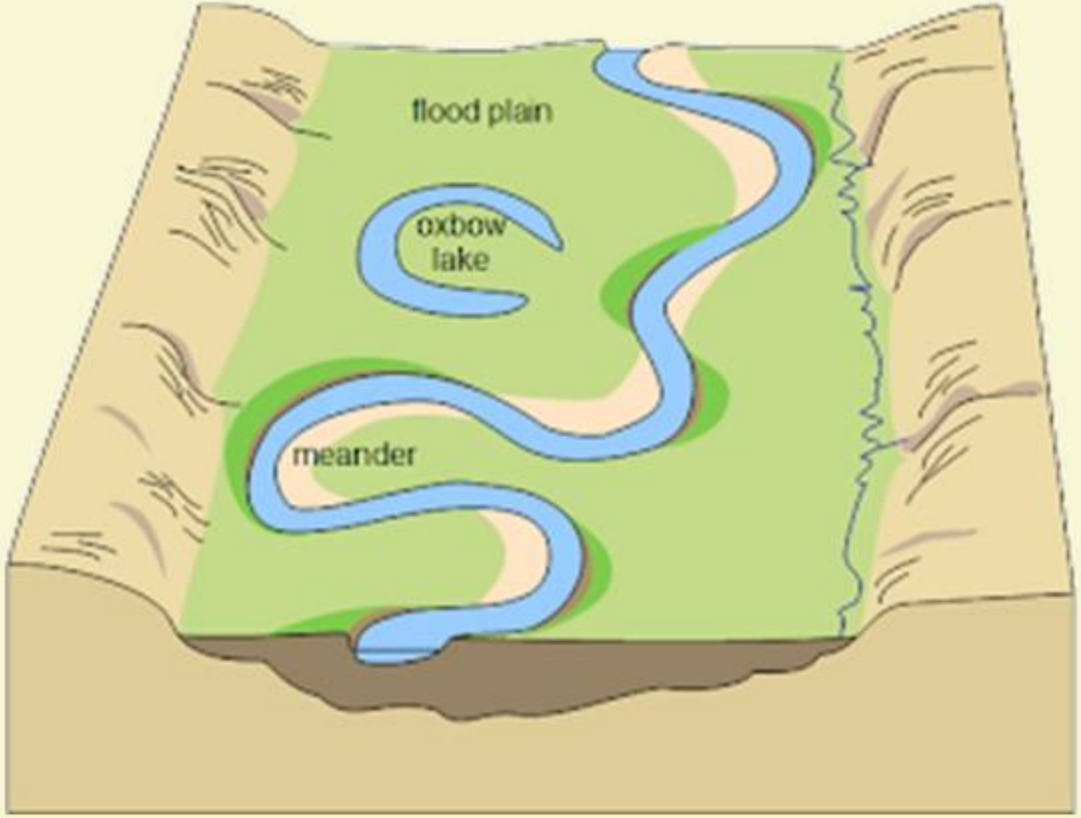


1. The river moving downstream across a broad, level plain is heavy with debris brought down from the upper course.
2. In this stage, vertical corrasion has almost ceased though lateral corrasion still goes on to erode its banks further. The work of the river is mainly deposition and forming extensive flood plains.
3. Coarse materials are dropped and the finer silt is carried downwards to the mouth of the river.

FLUVIAL LANDFORMS

THE COURSE OF RIVER

3 THE LOWER OR PLAIN COURSE



OX-BOW LAKES

4 Large sheets of materials are deposited on the flood plain and this may split the river into several complicated channels, which is known as a ***braided stream***.

5 Some of the major plain course features are the following.

a) Flood plain
b) Ox-bow lakes
c) Delta

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